**Example 8.6** The marks scored by 10 students in a class test are 25, 29, 30, 33, 35, 37, 38, 40, 44, 48. Find the standard deviation.

***Solution*** The mean of marks is 35.9 which is not an integer. Hence we take assumed mean, $A = 35$, $n = 10$.

| $x_i$ | $d_i = x_i - A$<br>$d_i = x_i - 35$ | $d_i^2$ |
|---|---|---|
| 25 | −10 | 100 |
| 29 | −6 | 36 |
| 30 | −5 | 25 |
| 33 | −2 | 4 |
| 35 | 0 | 0 |
| 37 | 2 | 4 |
| 38 | 3 | 9 |
| 40 | 5 | 25 |
| 44 | 9 | 81 |
| 48 | 13 | 169 |
| | $\Sigma d_i = 9$ | $\Sigma d_i^2 = 453$ |

Standard deviation

$$\sigma = \sqrt{\frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma d_i}{n}\right)^2}$$

$$= \sqrt{\frac{453}{10} - \left(\frac{9}{10}\right)^2}$$

$$= \sqrt{45.3 - 0.81}$$

$$= \sqrt{44.49}$$

$$\sigma \simeq 6.67$$

**(ii) Step deviation method**

Let $x_1, x_2, x_3, \ldots x_n$ be the given data. Let A be the assumed mean.

Let $c$ be the common divisor of $x_i - A$.

$$\text{Let} \quad d_i = \frac{x_i - A}{c}$$

$$\text{Then} \quad x_i = d_i c + A \qquad \ldots(1)$$

$$\Sigma x_i = \Sigma\left(d_i c + A\right) = c\Sigma d_i + A \times n$$

$$\frac{\Sigma x_i}{n} = c\frac{\Sigma d_i}{n} + A$$

$$\bar{x} = c\bar{d} + A \qquad \ldots(2)$$

$$x_i - \bar{x} = cd_i + A - c\bar{d} - A = c(d_i - \bar{d}) \quad \text{(using (1) and (2))}$$

$$\sigma = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n}} = \sqrt{\frac{\Sigma(c(d_i - \bar{d}))^2}{n}} = \sqrt{\frac{c^2\Sigma(d_i - \bar{d})^2}{n}}$$

$$\sigma = c \times \sqrt{\frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma d_i}{n}\right)^2}$$

**Note**

We can use any of the above methods for finding the standard deviation

**Activity 1**

Find the standard deviation of the marks obtained by you in all five subjects in the quarterly examination and in the midterm test separately. What do you observe from your results.



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